

JavaScript DOM Model

A Comprehensive Guide with
Examples

Adding JavaScript to a Website

- 1. Inline JavaScript:

```
<button onclick="alert('Hello!')">Click Me</button>
```

- 2. Internal JavaScript:

```
<script>  
  console.log("Hello, World!");  
</script>
```

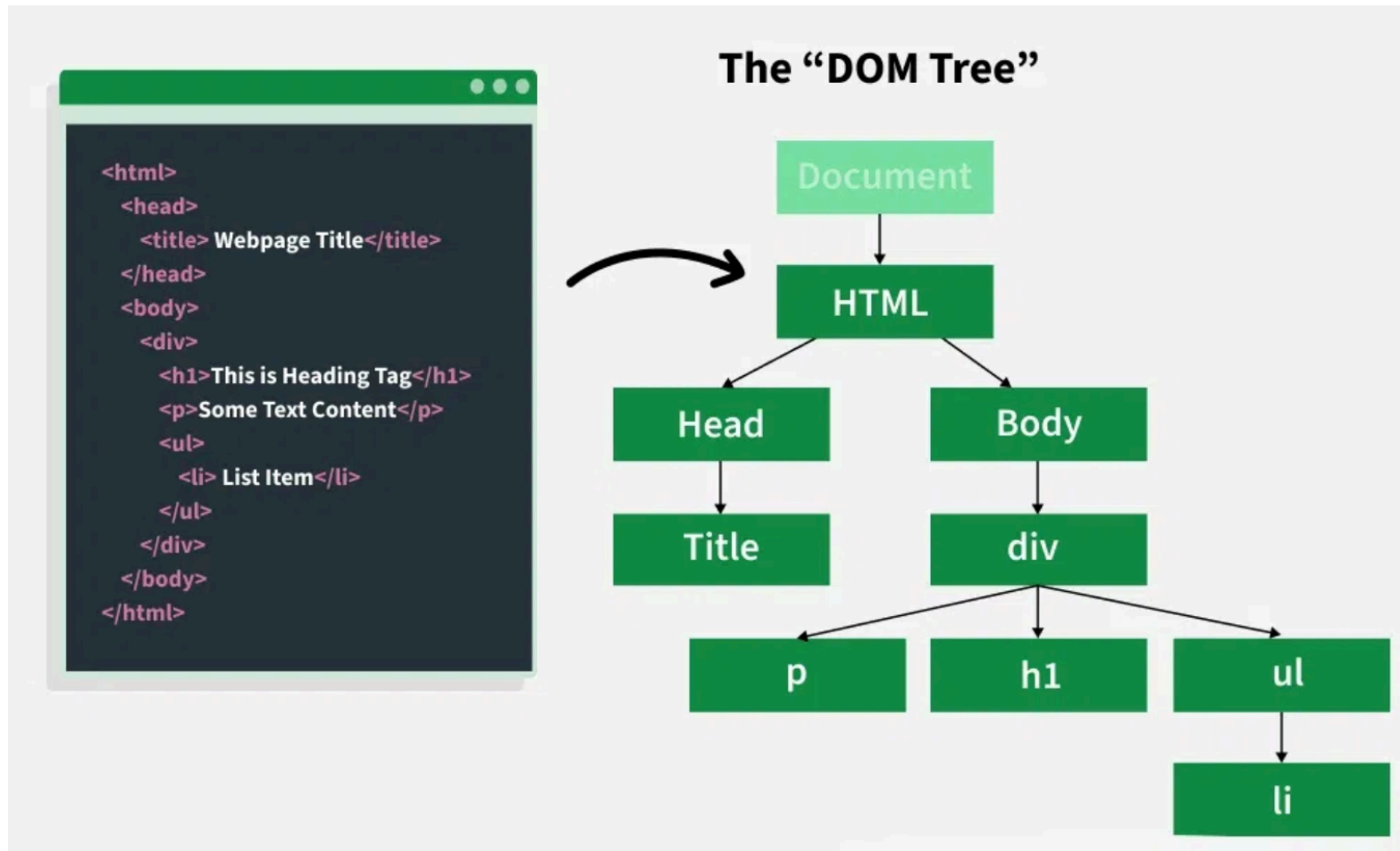
- 3. External JavaScript:

```
<script src="script.js"></script>
```

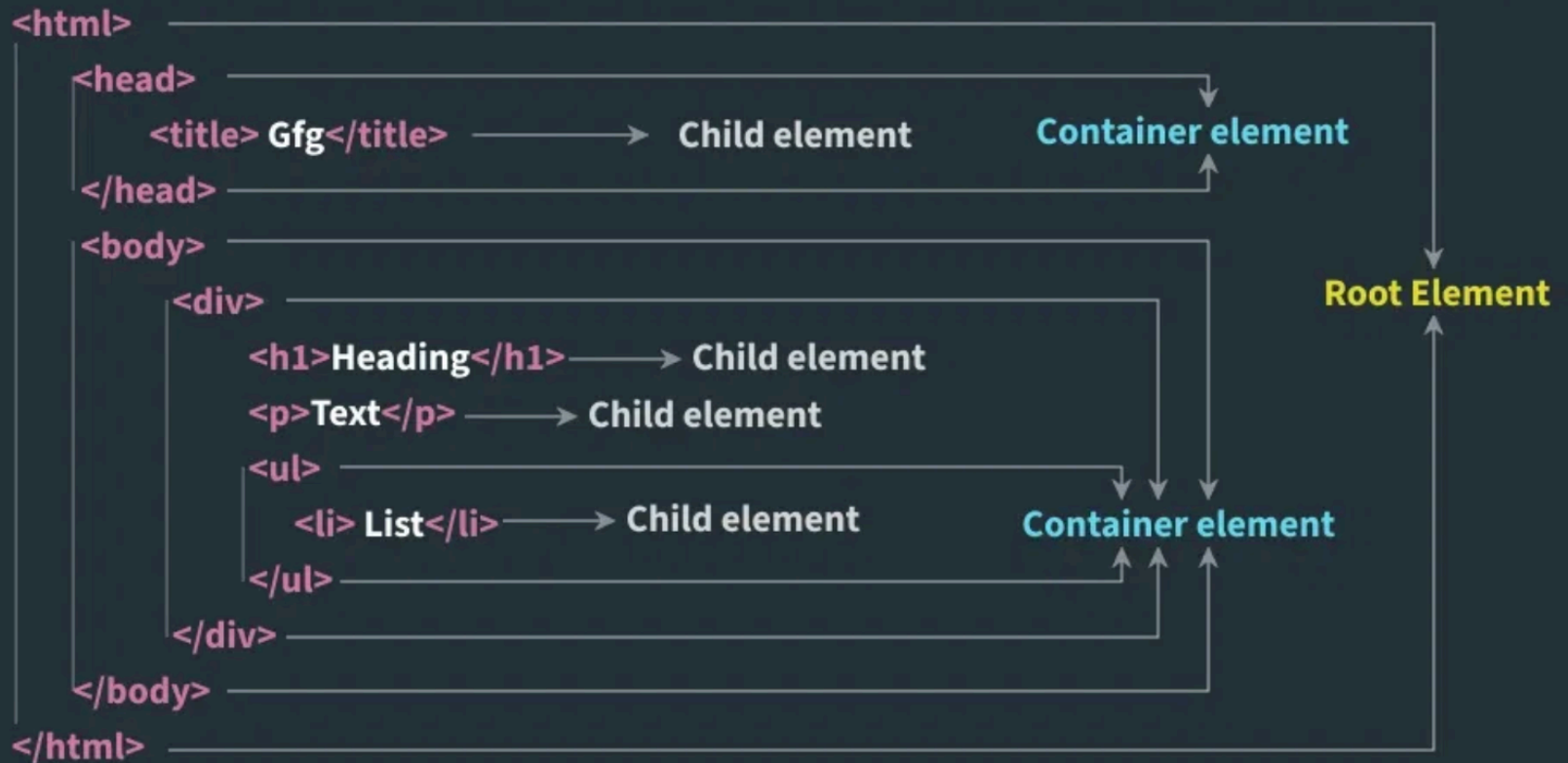
Introduction to the DOM

- The Document Object Model (DOM) is a programming interface for web documents.
- Represents the structure of an HTML document as a tree.
- Enables JavaScript to dynamically interact with web pages.

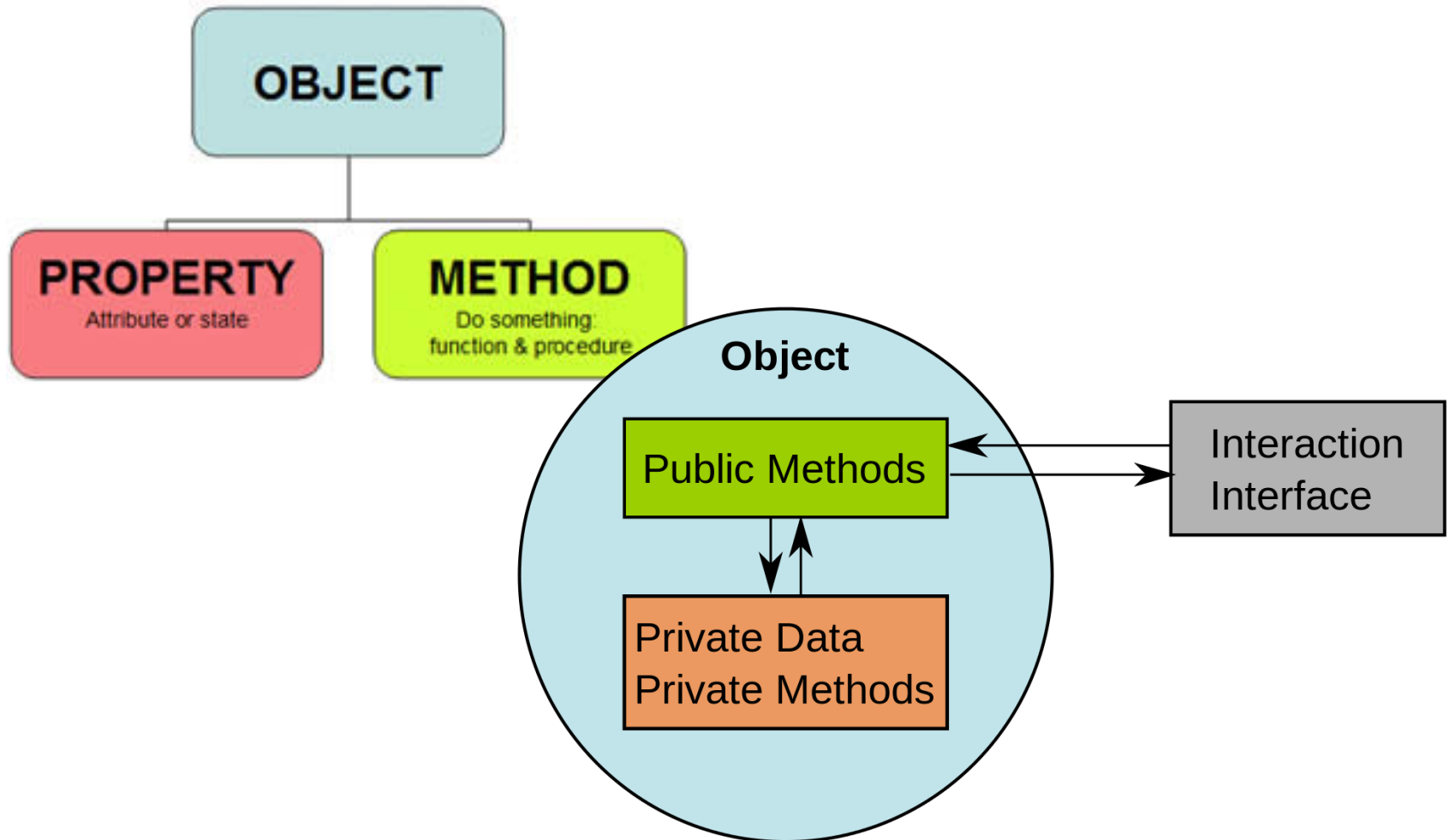
Document Object Model(DOM)



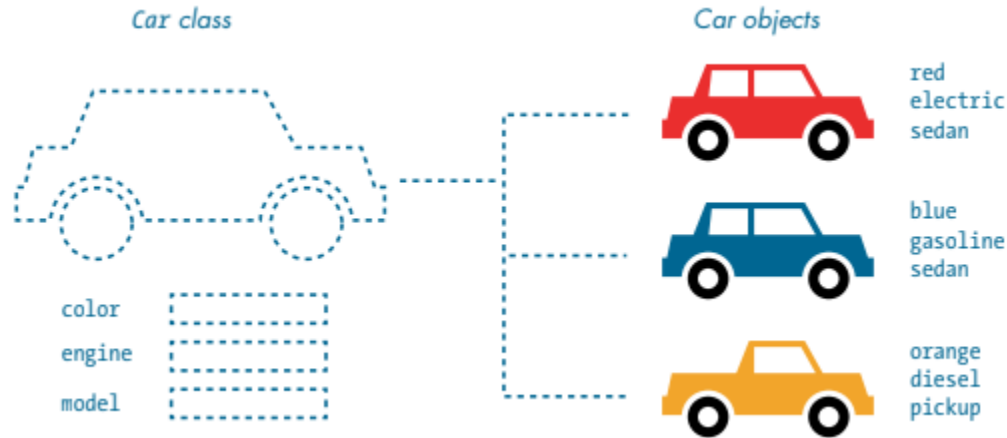
DOM



Objects and Encapsulation

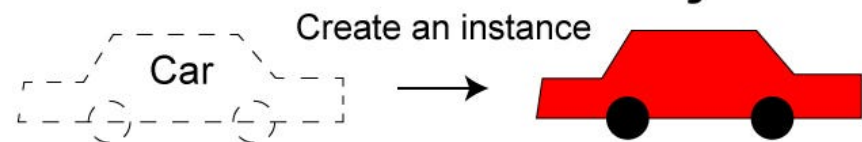


Objects: Properties and Method



Class

Object

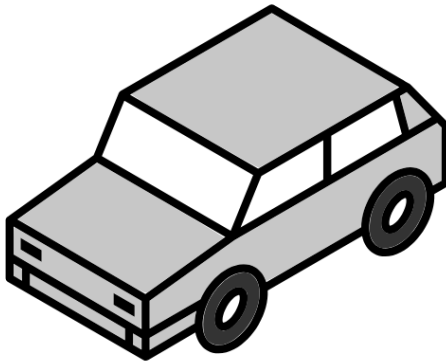


Properties	Methods - behaviors
color	start()
price	backward()
km	forward()
model	stop()

Property values	Methods
color: red	start()
price: 23,000	backward()
km: 1,200	forward()
model: Audi	stop()

Get and Set

Object



Properties

```
car.name="Mitsubishi"  
car.model="Super Lancer"  
car.weight=1300  
car.year=1996  
car.color="grey"
```

Methods

```
car.start()  
car.drive()  
car.break()  
car.honk()
```


JS:Object

- Button as Object
 - Properties
 - innerHTML
 - style
 - firstChild
 - Methods
 - click()
 - appendChild()
 - setAttribute()

Challenges

- Change the third list element to your name without any changes in HTML file.

1. DOM Object Properties

- Properties provide information about DOM elements and allow getting or setting their values.
- Common Properties:
 - **innerHTML** – Gets or sets the HTML content of an element.
 - **innerText** / **textContent** – Gets or sets the text content of an element.
 - **id** – Gets or sets the id of an element.
 - **className** – Gets or sets the class of an element.
 - **value** – Gets or sets the value of form elements (input, textarea).
 - **src** – Gets or sets the source of an image or iframe.
 - **href** – Gets or sets the link destination of an anchor tag.

DOM Object Properties

- Example:

```
let heading = document.getElementById("title");  
console.log(heading.innerText); // Get text content  
heading.innerText = "New Title"; // Set text content
```

2. DOM Object Methods

- Methods perform actions on elements, like selecting, modifying, or deleting them.
- **Common Methods:**
 - Selection Methods:
 - **getElementById("id")** – Selects an element by id.
 - **getElementsByClassName("class")** – Selects elements by class.
 - **getElementsByTagName("tag")** – Selects elements by tag name.
 - **querySelector("selector")** – Selects the first matching element.
 - **querySelectorAll("selector")** – Selects all matching elements.

querySelector()

- 1. Select by ID

```
// HTML  
// <div id="unique">This is a unique element</div>  
  
const elementById = document.querySelector("#unique");  
console.log(elementById.textContent); // Output: This is a unique e
```

querySelector()

- 2. Select by Class

```
// HTML  
// <div class="highlight">Highlighted Element</div>  
  
const elementByClass = document.querySelector(".highlight");  
console.log(elementByClass.textContent); // Output: Highlighted Ele
```

querySelector()

- 3. Select by Tag Name

```
// HTML
// <p>This is a paragraph</p>

const elementByTag = document.querySelector("p");
console.log(elementByTag.textContent); // Output: This is a paragraph
```

↓

querySelector()

- 4. Select with Attribute

javascript

 Copy code

HTML

```
<input type="text" placeholder="Enter text here">
```

```
st elementByAttribute = document.querySelector('input[type="text"]');  
sole.log(elementByAttribute.placeholder); // Output: Enter text here
```

querySelector()

- 5. Select with Descendant Combinator

```
// HTML
// <div class="container">
//   <span class="child">Child Element</span>
// </div>

const descendant = document.querySelector(".container .child");
console.log(descendant.textContent); // Output: Child Element
```

2. DOM Object Methods con...

– Manipulation Methods:

- **setAttribute("attribute", "value")** – Sets a new attribute value.
- **getAttribute("attribute")** – Gets an attribute value.
- **removeAttribute("attribute")** – Removes an attribute.

2. DOM Object Methods con...

– Content and Structure Methods:

- **appendChild(childElement)** – Adds a child element.
- **removeChild(childElement)** – Removes a child element.
- **replaceChild(newChild, oldChild)** – Replaces a child element.
- **createElement("tagName")** – Creates a new element.

– Example:

```
let newPara = document.createElement("p");  
newPara.innerText = "This is a new paragraph.";  
document.body.appendChild(newPara); // Add to the document
```

2. DOM Object Methods con...

– Event-Handling Methods:

- **addEventListener("event", function)** – Adds an event listener.
- **removeEventListener("event", function)** – Removes an event listener.

– Example:

```
document.querySelector("button").addEventListener("click", function  
    alert("Button clicked!");  
});
```

Challenges

- To do List challenge